

# Investigating Display Position of a Head-Fixed Augmented Reality Notification for Dual-task

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## ABSTRACT

Providing additional information in the proper position of augmented reality (AR) head-mounted display (HMD) can help increase AR performance and usability for dual-task. Therefore, our study investigated how to place notifications for the dual-task to address this. We compared eight display positions and two tasks (single and dual tasks) to identify the appropriate area for displaying notifications. We confirmed that the middle-right reduces response time and task load. In contrast, the top-left is the location, which should avoid providing any notification in AR dual-task. Our study contributes to designing AR notifications on HMDs to enhance everyday AR experiences.

**Index Terms:** Human-centered computing—Human Computer Interaction (HCI)—Interaction paradigms—Virtual reality, Mixed / augmented reality

## 1 INTRODUCTION

As the adoption of augmented reality (AR) technology increases in various fields, AR head-mounted displays (HMD) are envisioned to be used routinely like smartphones. The notification mechanism is one of the essential functions of a typical digital device in everyday use. However, AR HMDs are a new computing environment that is different from traditional smart devices to provide notifications. Therefore, it is necessary to understand the new method for registering AR notifications on the HMDs in everyday scenarios.

The AR HMDs make it to provide additional information directly superimposed on the user's field of view (FOV). However, the user has limited to a region to focus on [1], as well they are influenced by the complexity of real-world background to accept information [4]. Thus, if AR notification is displayed inappropriately on HMDs, users are possible to experience cognitive load or safety problems. The AR HMDs guidelines suggested that they are not recommended to use a head-fixed coordinate system when displaying virtual information. However, head-anchoring has the advantage of allowing users to respond immediately to virtual information [5]. Nevertheless, there is a lack of study considering how to place head-fixed notification in binocular AR HMDs in a dual-task situation.

In order to address this gap, we investigated where to place the AR notifications on HMDs. We compared eight different positions for two tasks. Participants were engaged in 1) a single task for detecting augmented notification and 2) a dual-task consisting of a sequence of notification tasks combining the watching TV task. It was possible to specify a suitable location for AR notification from this.

## 2 METHOD

The experiment was a  $2 \times 8$  within-subject design with two independent variables: display position (eight positions) and task (single, dual-task). For the single task, they provided notification tasks only.

On the other hand, for the dual-task, they have experienced watching TV tasks and notification tasks simultaneously. A sequence of the position (one block) was counter-balanced using Latin Square. We measured response time and perceived task load assessed by the Raw TLX (RTLX) [2] as dependent variables. The experiment was performed on only eight positions, excluding the middle-center. The stimuli were placed over the TV screen. The study consisted of eight blocks because we evaluated eight different positions. Each of the blocks consisted of 6 trials. In summary, participants have conducted 8 positions  $\times$  6 trials  $\times$  2 different tasks.

### 2.1 Setup

We ran an application using Magic Leap One. It was developed in Unity 2019.1.9f1 to provide notifications on the eight positions of the screen. We placed those notification stimuli using a head-fixed coordinate system. The experiment was conducted at a room size of  $3.3\text{m} \times 3.5\text{m}$  ( $11.55\text{m}^2$ ). The flat-screen TV (40 inches TV with  $1920 \times 1080$  resolution) was placed considering the recommended viewing distance to mimic a real-life. The videos provided to participants for the dual-task were one of the popular TV show's video clips on Youtube of the countries where the experiments were conducted. We created a list of 24 video clips (15-minute length) and randomly provided them from that list. The participants used a Bluetooth keyboard proceeded with the experiment to collect response time data. These data were collected by pressing either the up or down button on a keyboard. We also maintained uniform lighting conditions in the whole study. This study was approved by the Institutional Review Board (IRB).

### 2.2 Stimuli

We selected three elevations ( $+10^\circ$ ,  $0^\circ$ ,  $-10^\circ$ ) and three azimuths ( $+10^\circ$ ,  $0^\circ$ ,  $-10^\circ$ ) to control the same distance. The angle of  $10^\circ$  was set based on the limitation of the human's optimal upper region at the horizon [6]. Moreover, we selected a square shape as a notification stimulus to adjust the same distance from the center. The stimulus included a small white hole, either  $1^\circ$  above or below from the center of the stimulus. The positions of these holes (either above or below) are randomized so that the participants cannot predict its' position in advance. Each notification stimulus was displayed for 1000 milliseconds. Furthermore, the stimulus was activated at random intervals between 5 and 10 seconds to prevent participants from expecting when the following notification would appear. The stimulus's color was chosen to be gray, suitable for overlay on the video content [3]. Before the study, gray crosshair stimulus was displayed on each TV screen and HMD screen only at the beginning of the experiment to provide notification an accurate location by adjusting the user's line of sight. The crosshairs represent the center of the HMD's screen and TV screen. The stimuli were placed 1.5m in front of the user.

### 2.3 Procedure

We recruited 24 participants (10 males) through our university website. Their age range was from 20 to 31 years ( $M=24.75$ ,  $SD=3.18$ ). All participants had a normal or corrected-to-normal vision over 0.8 degrees of their sight. Before the main study, we instructed participants how to input the time data using a keyboard and fitted the

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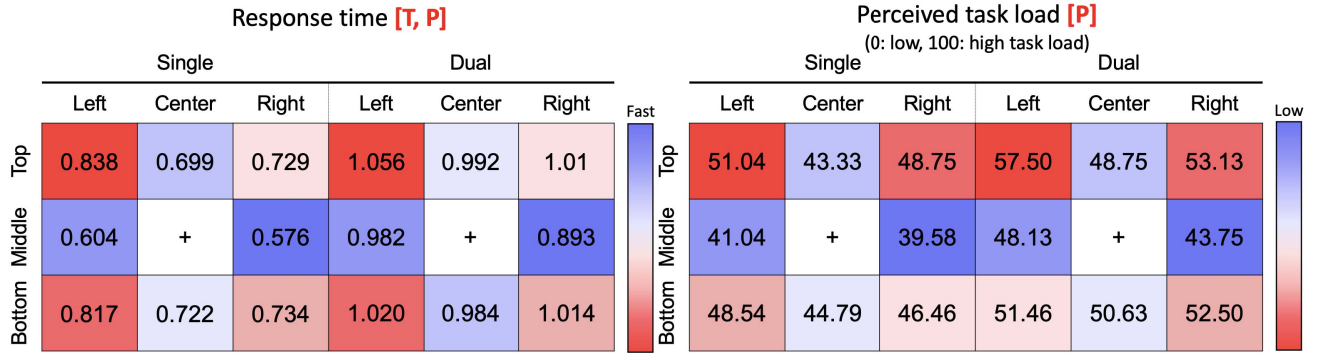


Figure 1: Response time and perceived task load results. Red T and P in the square bracket: a significant effect of Task and Position; I: significant Interaction effect. The unit of measurement: response time (second), and perceived task load (average score).

headset on their heads. Next, we asked them to sit on a chair in front of the TV. Then, when the gray crosshairs were registered on each screen (TV and HMDs screen, respectively), the participant looked straight ahead and adjusted the center by aligning the crosshairs on the TV screen with the crosshairs on the HMD. For the dual-task condition, we asked participants to watch a video through the TV. Participants experienced eight display positions in the two different tasks. We confirmed that the participant understands what the content of the notification by entering the white square hole positions (up or down). Each block was based on one position, so each task conducted eight blocks. After performing one block for each position, we provided the questionnaires to reduce forgetting about the results of the experimental experience. Participants took about one hour for the experiment. They received 10 dollars per hour for conducting our study.

### 3 RESULT

We performed statistical data analysis with SPSS 25. For the non-parametric data, we executed the ART (Aligned Rank Transform) using the ARTool toolkit [7]. To test for main and interaction effects, we conducted repeated-measures ANOVAs ( $\alpha = .05$ ). Mauchly's test was conducted to confirm the assumption of sphericity. We used pairwise comparisons as a post hoc test, and Bonferroni correction was applied to the p-values.

**Response Time** We found the main effects of positions ( $F(7,161)=7.032, p<.001, \eta_p^2=.234$ ) and tasks ( $F(1,23)=5.195, p<.05, \eta_p^2=.184$ ). In a pairwise comparison of tasks, participants detected the AR notification faster while conducting the simple task ( $p<.05$ ). Moreover, the post hoc analysis with pairwise comparison showed that the middle-right ( $M=.735, SD=.458$ ) was faster than the other positions for the position factor (all  $p<.05$ ). We also found that top-left ( $M=.947, SD=.490$ ) was significantly slower than the other positions (all  $p<.01$ ). Furthermore, the bottom-center ( $M=.853, SD=.387$ ) was faster than the top-right ( $M=.870, SD=.489; p<.05$ ). Participants detected faster on bottom-right than on bottom-left ( $M=.919, SD=.529; p<.05$ ). However, there was no significant interaction between positions and tasks ( $F(7,161)=1.100, p=.366, \eta_p^2=.045$ ).

**Perceived Task Load** A main effect of position was found ( $F(7,161)=19.227, p<.001, \eta_p^2=.455$ ). Participants found that middle-left had a lower perceived task load than top-left, top-right, bottom-left, and bottom-right (all  $p<.05$ ). The middle-right was also lower than the four above positions (all  $p<.01$ ). Furthermore, the top-left had the highest perceived task load than the other locations (all  $p<.001$ ). However, there were no significant

effects for task ( $F(1,23)=1.599, p=.219, \eta_p^2=.065$ ) and interaction ( $F(4,118,94.713)=.789, p=.597, \eta_p^2=.033$ ).

### 4 CONCLUSION

We investigated display positions of AR HMDs and how they affect AR notification performance and experience while watching TV. Thus, we compared the effect of two types of tasks and eight positions for notification experience. Our findings supported that the middle-right position was reduced response time and perceived task load. On the other hand, top-left should be unwilling to provide any AR notification in dual-task. These findings can be used to design AR notifications on AR HMDs. Therefore, this could help designers and developers upbuild practical notification features for AR everyday usage scenarios.

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